

LC 1335: Min Difficulty of a Job Schedule

Targeted by: SAP Labs, Adobe

Problem Rules & Conditions

- Want to schedule a list of jobs in d days.
- To work on the i^{th} job, all j jobs need to be done where $0 \leq j < i$.
- Finish **at least one task** every day.

Key Definitions

- **Difficulty of a day** $\rightarrow$ max difficulty of a job done on that day.
- **Difficulty of a schedule** $\rightarrow \Sigma$ difficulty of each day for d days.

Input & Output

- **Given:** d , jobdifficulty array
- **O/P:** min difficulty of a job schedule

Basically, here d is acting like partitions.
i.e., if $d == 3$ we need to divide into 3 parts.

STANDARD PARTITION DP PROBLEM

C++ Implementation

```
class Solution {
public:
    int minDifficulty(vector<int>& jobdiff, int d) {

        int n=jobdiff.size();
        if(d>n) return -1;

        vector<vector<int>> dp(n+1,vector<int>(d+1,INT_MAX));
        dp[0][0]=0; //0 kaam ko 0 days me krne ke 0 tareeke

        for(int i=1;i<=n;i++){
            for(int parts=1;parts<=d;parts++){

                for(int j=parts-1;j<i;j++){

                    int lastmaxu=0;
                    for(int z=j;z<i;z++){
                        lastmaxu=max(lastmaxu,jobdiff[z]);
                    }

                    if(dp[j][parts-1]==INT_MAX) continue;
                    int curr=dp[j][parts-1]+lastmaxu;
                    dp[i][parts]=min(dp[i][parts],curr);
                }
            }
        }

        return dp[n][d];
    }
};
```